

```

In[*]:= (*TOMA DE DATOS*)

In[*]:= (*CALENTAMIENTO*)

In[*]:= VCal = {0.4, 0.47, 0.54, 0.62, 0.70, 0.79, 0.89, 1.00, 1.11, 1.24,
               1.36, 1.51, 1.66, 1.95, 2.11, 2.30, 2.50, 2.71, 2.93, 3.14}; (*mV*)

In[*]:= PCal = VCal / 0.16; (*mW*)

In[*]:= TCal = {100, 110, 120, 130, 140, 150, 160, 170, 180, 190,
               200, 210, 220, 240, 250, 260, 270, 280, 290, 300} + 273.15; (*K*)

In[*]:= dataCal = Transpose[{(TCal^4) / 10^10, PCal}];
               \[transposición\]

In[*]:= (*ENFRIAMIENTO*)

In[*]:= VEnf = {2.93, 2.75, 2.58, 2.41, 2.26, 2.12,
               1.97, 1.83, 1.69, 1.54, 1.41, 1.28, 1.15, 1.04, 0.91}; (*mV*)

In[*]:= PEnf = VEnf / 0.16; (*mW*)

In[*]:= TEnf = {290, 280, 270, 260, 250, 240, 230, 220, 210, 200, 190, 180, 170, 160, 150} + 273.15;
               (*K*)

In[*]:= dataEnf = Transpose[{(TEnf^4) / 10^10, PEnf}];
               \[transposición\]

In[*]:= (*ANÁLISIS DE DATOS*)

In[*]:= (*CALENTAMIENTO*)

In[*]:= fitCal = NonlinearModelFit[dataCal, A + B * x, {A, B}, x]
               \[ajusta a modelo no lineal\]

Out[*]=

FittedModel[-1.24 + 1.94 x]

In[*]:= fitCal[{"BestFit", "ParameterTable"}]

Out[*]=


|                         |   | Estimate | Standard Error | t-Statistic | P-Value                   |
|-------------------------|---|----------|----------------|-------------|---------------------------|
| { -1.23569 + 1.93876 x, | A | -1.23569 | 0.0293242      | -42.1388    | $1.92304 \times 10^{-19}$ |
|                         | B | 1.93876  | 0.00480036     | 403.878     | $4.4958 \times 10^{-37}$  |



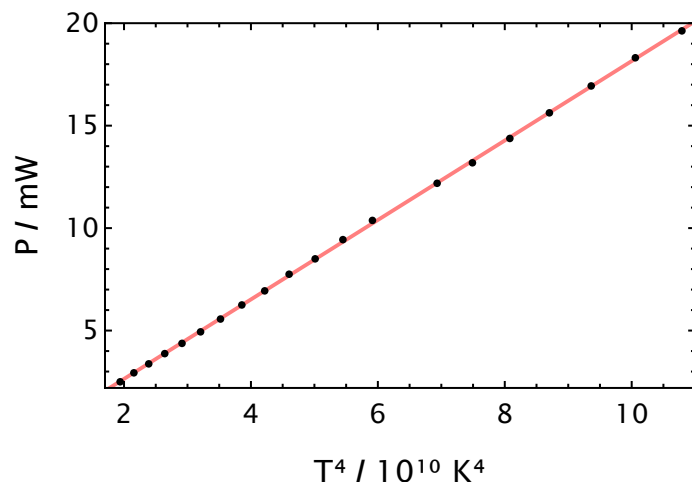
In[*]:= plotfitCal =
  Plot[fitCal[x], {x, 0, 30}, BaseStyle -> {FontFamily -> "CambriaMath", FontSize -> 14},
  \[representación gráfica\] \[estilo base\] \[familia de tipo de letra\] \[tamaño de tipo de letra\]
  PlotStyle -> {Blend[{Red, White}], Thick},
  \[estilo de representación\] \[mezcla\] \[rojo\] \[blanco\] \[grueso\]
  PlotRange -> {{1.7, 11}, {2.2, 20}}, Frame -> True,
  \[rango de representación\] \[marco\] \[verdadero\]
  FrameLabel -> {Style["T^4 / 10^10 K^4", Bold, 16], Style["P / mW", 16, Bold]};
  \[etiqueta de marco\] \[estilo\] \[negrita\] \[estilo\] \[negrita\]

In[*]:= plotdataCal = ListPlot[dataCal, PlotStyle -> {Black, Thickness[0.003]};
               \[representación de lista\] \[estilo de representación\] \[negro\] \[grosor\]

```

```
In[*]:= Show[plotfitCal, plotdataCal]
```

```
Out[*]=
```



```
In[*]:= (*ENFRIAMIENTO*)
```

```
In[*]:= fitEnf = NonlinearModelFit[dataEnf, A + B * x, {A, B}, x]
```

```
Out[*]=
```

```
FittedModel[0.318+1.83 x]
```

```
In[*]:= fitEnf[{"BestFit", "ParameterTable"}]
```

```
Out[*]=
```

```
{0.318127 + 1.82536 x, 

|   | Estimate | Standard Error | t-Statistic | P-Value                     |
|---|----------|----------------|-------------|-----------------------------|
| A | 0.318127 | 0.212077       | 1.50005     | 0.15749                     |
| B | 1.82536  | 0.0324325      | 56.2818     | 6.48166 × 10 <sup>-17</sup> |

}
```

```
In[*]:= plotfitEnf =
```

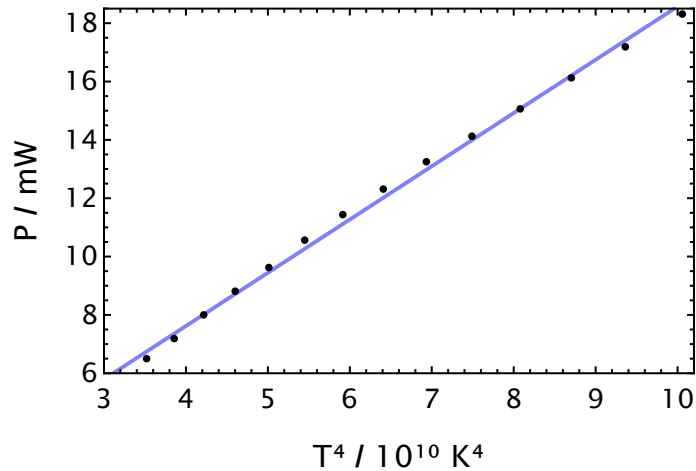
```
Plot[fitEnf[x], {x, 0, 40}, BaseStyle -> {FontFamily -> "CambriaMath", FontSize -> 14},
PlotStyle -> {Blend[{Blue, White}], Thick},
PlotRange -> {{3, 10.2}, {6, 18.5}}, Frame -> True,
FrameLabel -> {Style["T^4 / 10^10 K^4", Bold, 16], Style["P / mW", 16, Bold]}];
```

```
In[*]:= plotdataEnf = ListPlot[dataEnf, PlotStyle -> {Black, Thickness[0.003]}];
```

```
In[*]:= Show[plotfitEnf, plotdataEnf]
```

muestra

```
Out[*]=
```



```
In[*]:= (*ln(P+P0)*)
```

```
In[*]:= (*CALENTAMIENTO*)
```

```
In[*]:= dataCal2 = Transpose[{Log[TCal], Log[PCal + 1.2356884207856167]}];
```

transposición logaritmo logaritmo

```
In[*]:= fitCal2 = NonlinearModelFit[dataCal2, A + B * x, {A, B}, x]
```

ajusta a modelo no lineal

```
Out[*]=
```

FittedModel[$-22.4 + 4.01x$]

```
In[*]:= fitCal2[{"BestFit", "ParameterTable"}]
```

```
Out[*]=
```

	Estimate	Standard Error	t-Statistic	P-Value
$-22.4365 + 4.01174x$, A	-22.4365	0.0558747	-401.55	4.98885×10^{-37}
B	4.01174	0.00908676	441.492	9.05246×10^{-38}

```
In[*]:= plotfitCal2 =
```

```
Plot[fitCal2[x], {x, 0, 30}, BaseStyle -> {FontFamily -> "CambriaMath", FontSize -> 14},
```

representación gráfica estilo base familia de tipo de letra tamaño de tipo de letra

```
PlotStyle -> {Blend[{Red, White}], Thick},
```

estilo de representación mezcla rojo blanco grueso

```
PlotRange -> {{5.9, 6.36}, {1.3, 3.1}}, Frame -> True,
```

rango de representación marco verdadero

```
FrameLabel -> {Style["Log[T / K]", Bold, 16], Style["Log[P+P. / mW]", 16, Bold]}];
```

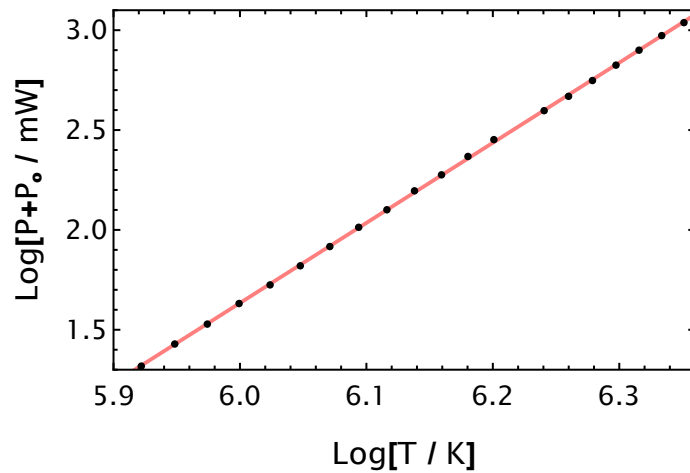
etiqueta de marco estilo logaritmo negrita estilo logaritmo negrita

```
In[*]:= plotdataCal2 = ListPlot[dataCal2, PlotStyle -> {Black, Thickness[0.003]}];
```

representación de lista estilo de representación negro grosor

```
In[*]:= Show[plotfitCal2, plotdataCal2]
```

```
Out[*]=
```



```
In[*]:= (*ENFRIAMIENTO*)
```

```
In[*]:= dataEnf2 = Transpose[{Log[TEnf], Log[PEnf - 0.31812694686808046]}];
```

```
In[*]:= fitEnf2 = NonlinearModelFit[dataEnf2, A + B * x, {A, B}, x]
```

```
Out[*]=
```

```
FittedModel[ -23.3+4.14 x ]
```

```
In[*]:= fitEnf2[{"BestFit", "ParameterTable"}]
```

```
Out[*]=
```

	Estimate	Standard Error	t-Statistic	P-Value
-23.2822 + 4.13798 x, A	-23.2822	0.530756	-43.8662	1.62866×10^{-15}
B	4.13798	0.0856392	48.3188	4.66785×10^{-16}

```
In[*]:= plotfitEnf2 =
```

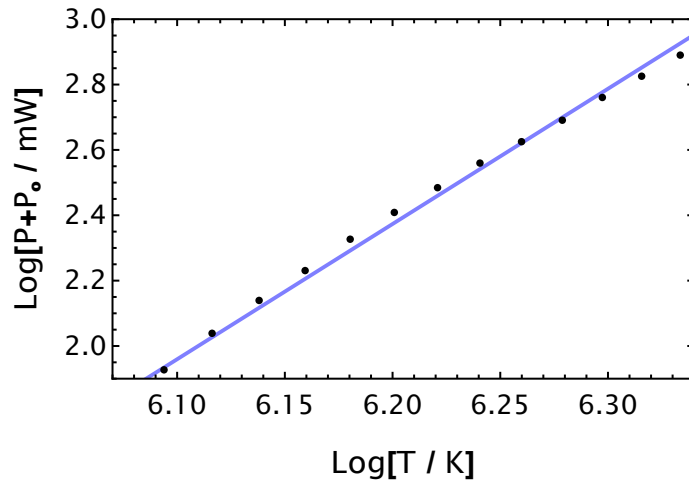
```
Plot[fitEnf2[x], {x, 0, 30}, BaseStyle -> {FontFamily -> "CambriaMath", FontSize -> 14},
PlotStyle -> {Blend[{Blue, White}], Thick},
PlotRange -> {{6.07, 6.34}, {1.9, 3.0}}, Frame -> True,
FrameLabel -> {Style["Log[T / K]", Bold, 16], Style["Log[P+P₀ / mW]", 16, Bold]};
```

```
In[*]:= plotdataEnf2 = ListPlot[dataEnf2, PlotStyle -> {Black, Thickness[0.003]}];
```

```
In[*]:= Show[plotfitEnf2, plotdataEnf2]
```

muestra

```
Out[*]=
```



```
In[*]:= (*OTROS CÁLCULOS*)
```

```
In[*]:= Sprime = Pi * (12.5) ^ 2; (*mm^2*)
```

número pi

```
ra = 10; (*mm*)
```

```
In[*]:= d = 120; (*mm*)
```

```
In[*]:= f = (Sprime * ra ^ 2 / (d ^ 2)) * 10 ^ (-6) (*m^2*)
```

```
Out[*]=
```

3.40885×10^{-6}

```
In[*]:= Errrf = (2 * f * 1.4 / d) (*m^2*)
```

```
Out[*]=
```

7.95397×10^{-8}

```
In[*]:= SigmaTab = 5.670374 * 10 ^ -8;
```

```
In[*]:= (*SIGMA CALENTAMIENTO*)
```

```
In[*]:= SigmaCal = 1.938762432097837 * 10 ^ -13 / f (*W/m2 K4*)
```

```
Out[*]=
```

5.68744×10^{-8}

```
In[*]:= ErrSigmaCal =
```

```
Sqrt[(0.004800362231865056 * 10 ^ -13 / f) ^ 2 + (1.938762432097837 * 10 ^ -13 * Errrf / f ^ 2) ^ 2]
```

raíz cuadrada

```
Out[*]=
```

1.40821×10^{-10}

```
In[*]:= DifRelCal = (Abs[SigmaTab - SigmaCal] / SigmaTab) * 100
```

valor absoluto

```
Out[*]=
```

0.301051

```
In[*]:= (*SIGMA ENFRIAMIENTO*)
```

```

In[*]:= SigmaEnf = 1.8253596131651195 * 10^-13 / f
Out[*]=

$$5.35477 \times 10^{-8}$$


In[*]:= ErrSigmaEnf =
  Sqrt[(0.03243252895948919 * 10^-13 / f)^2 + (1.8253596131651195 * 10^-13 * Errf / f^2)^2]
  |raíz cuadrada
Out[*]=

$$9.51422 \times 10^{-10}$$


In[*]:= DifRelEnf = (Abs[SigmaTab - SigmaEnf] / SigmaTab) * 100
  |valor absoluto
Out[*]=
5.5658

In[*]:= (*h Planck con CALENTAMIENTO*)
In[*]:= h = ((2 * (Pi^5) * ((1.380658 * 10^(-23))^4)) / (15 * (299792458)^2 * SigmaCal))^(1/3)
  |número pi
Out[*]=

$$6.61949 \times 10^{-34} \left(4\right)^{1/3}$$


In[*]:= htab = 6.62607015 * 10^(-34);
In[*]:= DifRelh = (Abs[htab - 6.6194918730196354*10^-34] / htab) * 100
  |valor absoluto
Out[*]=
0.0992787

In[*]:= (*T0 con calentamiento*)
T0Cal = (1.2356884207856167 * 10^(-3) / (f * SigmaCal))^(1/4) - 273.15
Out[*]=
9.40051

```